

TROUBLE SHOOTING AND HELP DESK

TASK: 1

List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Five Common Desktop PC Hardware Issues

Hardware Issue	Typical Symptom
1. Faulty RAM	PC may fail to boot, show no display, restart repeatedly, or produce beep codes.
2. Faulty Power Supply (PSU)	PC may not turn on, randomly shut down, or restart unexpectedly.
3. Overheating CPU	PC may become very slow, shut down suddenly, or restart during heavy use.
4. Faulty/Loose Storage Drive	Operating system may fail to boot, files may become inaccessible, or the drive may not be detected.
5. Display/GPU Problem	Monitor may show a blank screen, no signal, or graphical errors.

TASK: 2

Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Laptop Not Charging – Troubleshooting Plan

Step 1: Check the Power Outlet

First, I would connect another working device to the same wall socket to confirm that the electrical outlet is working.

Step 2: Inspect the Charger

I would visually inspect the charger, cable, plug, and connector for cuts, burns, bent pins, or loose connections.

Step 3: Check the Charger Output

Using a multimeter set to **DC voltage**, I would check the voltage output of the laptop adapter according to the voltage printed on its label. A very low or zero reading could indicate a faulty charger.

Step 4: Check the Charging Connector

I would inspect the laptop's charging port for dust, damage, looseness, or bent contacts. I would also check whether the connector fits firmly.

Step 5: Check the Battery Connection

After disconnecting power and following safe servicing procedures, I would check whether the internal battery connector is properly connected and whether there is any visible damage.

Step 6: Check for Physical Damage

I would inspect the motherboard area around the charging circuit for burnt components, damaged connectors, or other visible signs of damage.

Step 7: Test With a Known-Good Compatible Charger

If available, I would test the laptop with a known-good charger having the **correct voltage, connector, and required current rating**. If it charges with the replacement, the original charger is likely faulty.

Conclusion

If the charger has the correct output but the laptop still does not charge, the problem could be related to the charging port, battery, charging circuit, or motherboard and may require professional repair.

TASK: 3

Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Testing a USB Phone Charger With a Multimeter

Equipment Used

- Digital multimeter
- USB phone charger
- USB cable/breakout connection suitable for measurement

Procedure

1. Set the multimeter to **DC voltage (V =)**.
2. Connect the charger to the power supply.
3. Identify the USB **5V/VBUS** and **GND** connections.
4. Place the black probe on GND and the red probe on the 5V/VBUS connection.
5. Read the voltage shown on the multimeter.

Recorded Reading

Measured voltage: 5.08 V DC

The expected USB voltage range given in the question is **4.75 V to 5.25 V**.

Result: 5.08 V is within the expected range, so the charger output is within the specified range.

TASK: 4

A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?
Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Testing the PSU 12V Rail With a Multimeter

Procedure

1. Switch off the PC and disconnect the AC power.
2. Disconnect the ATX power connector from the motherboard.
3. Make sure the PSU is handled safely and that no motherboard components are connected to the connector being tested.
4. Set the multimeter to **DC voltage (V=)**.
5. Identify the **yellow wire (+12 V)** and **black wire (ground)**.
6. With the PSU appropriately powered for the measurement, place the black probe on the black ground wire.
7. Place the red probe on the yellow +12 V wire.
8. Read and record the voltage displayed by the multimeter.

Recorded Reading

Measured 12V rail: 12.08 V DC

The reading is above **11.5 V**, so it does not indicate the low-voltage condition described in the question.

If the Reading Is Below 11.5 V

If the +12 V rail measures below **11.5 V**, it could indicate a **faulty or failing power supply** or an abnormal power-delivery problem. A weak PSU may cause random shutdowns, instability, or restarting under load.